



College of Computing and Informatics
Department of Computer Science

Course Syllabus

Course Title
Topics in CS1
(Introduction to Machine Learning)

Course Code
1501490

Fall, 2025 / 2026

Course Syllabus

Course Title	Topics in CS1 (Introduction to Machine Learning)		
Course Code	1501490		
Credit Hours	3		
Pre-requisite(s)	Knowledge in programming (Python)		
Co-requisite(s)	None		
Semester	Fall	Year	2025/2026
Instructors Name	Dr. Bouziane Brik		
Office Location	M5-207		
Tel. No.			
Email	bbrik@sharjah.ac.ae		
Lecture Times	Tuesday and Thursday 15:30 – 16:45		
Office Hours			

Course Description (as in the catalogue):

This course provides an introduction on machine learning. It gives an overview of many concepts, techniques and algorithms in machine learning such as supervised, unsupervised, and reinforcement learning. The course contents include **introduction to AI and Machine Learning**: definition, types, applications, etc. **Introduction to Python programming language**: Variables, Input/output instructions, selection and repetitive control structures, functions, libraries: math, NumPy, and Pandas. **Regression**: linear, multiple linear, polynomial, etc. **Classification**: logistic regression, KNN, Naïve-Bayes, and decision tree algorithm. **Unsupervised Learning**: K-means and DBSCAN algorithms. **Reinforcement Learning**: Q-Learning algorithm.

Course Objectives/Goals:

The goals of this course are to:

1. Learn and understand the most common machine learning techniques for unsupervised and supervised tasks.
2. Learn how to solve a problem using machine learning techniques.

Course Learning Outcomes:

By the end of successful completion of this course, the student will be able to:

- A. Analyze/process data related to a given problem.
- B. Use machine learning algorithms to train learning models on top of data.
- C. Design the suitable machine/deep learning algorithm.
- D. Implement machine/deep learning algorithms to train learning models.
- E. Validate the trained learning models using different metrics.

Alignment of Course Student Learning Outcomes to Program Student Learning Outcomes

Computer Science Program SLOs		Course SLOs
Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions.	Analyze	- Analyze/process data related to a given problem.
	Apply	- Use machine learning algorithms to train learning models on top of data.
Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.	Design	- Design the suitable machine learning algorithm.
	Implement	- Implement machine learning algorithms to train learning models.
	Evaluate	- Validate the trained learning models using different metrics.
Communicate effectively in a variety of professional contexts.		-
Describe professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.		-
Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.		-
Apply computer science theory and software development fundamentals to produce computing-based solutions.	Theory	- Process data and select the suitable machine learning algorithms.
	Development	- Implement machine learning algorithms to train learning models.

IT Multimedia Program SLOs		Course SLOs
Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions.	Analyze	- Analyze/process data related to a given problem.
	Apply	- Use machine learning algorithms to train learning models on top of data.
Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.	Design	- Design the suitable machine learning algorithm.
	Implement	- Implement machine learning algorithms to train learning models.
	Evaluate	- Validate the trained learning models using different metrics.

IT Multimedia Program SLOs		Course SLOs
Communicate effectively in a variety of professional contexts.		-
Describe professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.		-
Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.		-
Identify and analyze user needs and to take them into account in the selection, creation, integration, evaluation, and administration of computing-based systems.	Identify / Analyze	- Process and analyze input data and deciding about the suitable machine learning to apply
	Development	- Implement machine learning algorithms to train learning models.

Weekly Distribution of Course Topics/Contents

Week	Topic	Comments	Course SLO
1	Introduction to AI and Machine Learning: Definition, Types, applications, etc.	Introduce the students to AI and machine learning.	A, B
2	Introduction to Python programming language: Variables, Input/output instructions, selection and repetitive control structures, functions, libraries: math, NumPy, and Pandas.	Introduce the students to regression.	C, D, E
3	Regression: Linear, Multiple linear, and Polynomial.		
4	Regression: Linear, Multiple linear, and Polynomial. Quiz #1 Project		
5	Classification: Logistic regression and Naïve-Bayes, SVM, and KNN.	Introduce the students to classification	C, D, E
6	Classification: Logistic regression and Naïve-Bayes. Quiz #2		
7	Decision Tree: Algorithm, Implementation, and Evaluation.	Introduce the students Decision Tree algorithm.	C, D, E
8	Midterm exam		
9	Unsupervised Learning: Introduction, K-means algorithm	Introduce the students to random forest algorithm.	C, D, E
10	Unsupervised Learning: DBSCAN algorithm.		

11	Quiz #3	Introduce the students to unsupervised learning.	
12			
13	Reinforcement Learning: Q-Learning algorithm. Quiz #4 Due Date of project	Introduce the students to reinforcement learning	
14			
15	Final Exam		

Students' Assessment:

Students are assessed as follows:

Assessment Tool(s)	Date	Weight (%)
Project	Throughout the semester	20
Midterm	Midterm exams period in week7	25
Quizzes	Throughout the semester	10
Final Exam	Final exams period	45
Total		100

Course Outcome Assessment Plan:

Course Los	Teaching/Learning Method(s)	Assessment Tool(s)	Performance Indicators
Understand the basic operation principles of Machine Learning.	- Live Online Lectures / Recordings - Problem solving	- Quizzes - Exams - Project	Course assessment
Know how to apply different classes of machine learning algorithms.	- Live Online Lectures / Recordings - Problem solving	- Quizzes - Exams - Project	Course assessment
Solve the decision-making problems from different organizations, integrating intelligent tools.	- Live Online Lectures / Recordings Problem solving	- Quizzes - Exams - Project	Course assessment

Teaching and Learning Resources:

Python programming language.

Text Book(s):

- *Machine learning* - Mitchell, T.M, The McGraw-Hill Companies, 1997. ISBN: 0070428077.
- *Any Book on Machine Learning will do.*

Recommended Readings:

- Any available textbooks in the library about machine learning.

Attendance policy:

Attendance is compulsory. A student missing 10% (5 hours) of the total allocated course hours will receive 1st warning notice and a student missing 15% (7 hours) will receive 2nd warning notice. A student missing 20% (9 hours or more) will be forced to withdraw (in accordance with the university regulations).

Plagiarism/Cheating:

Students are expected to do their own work. You are allowed to work on project in teams only if specified by the instructor. In other words, students are encouraged to communicate about general principles of the course, but all assigned homework/project must be done on an individual basis. The instructor is available to provide any assistance that you may need. Cheating is considered a serious offense by the university. You should be aware of the severe penalty for cheating (refer to the student code of conduct published in the university catalogue).

Practical Aspect of the course:

Students are expected to practice main concepts of the course by writing computer programs in Python programming language.

Assessment and Learning Methods:

The course includes two lectures per week for 15 weeks, 3-4 quizzes (no makeup), and one project. The project were designed to enforce the problem solving and critical thinking skills.